

Dual Conic Proxies for AC-Optimal Power Flow



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AC-OPF (Optimal Power Flow)

$$\begin{aligned}
 \min \quad & \sum_{i \in \mathcal{N}} c_i \mathbf{p}_i^g \\
 \text{s.t.} \quad & \mathbf{S}_i^g - \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{S}}_e - \sum_{e \in \mathcal{E}_i^-} \tilde{\mathbf{S}}_e - (Y_i^s)^* |\mathbf{V}_i|^2 = \boxed{\mathbf{S}_i^d} \quad \forall i \in \mathcal{N} \quad \text{Supply-demand balance} \\
 & \vec{\mathbf{S}}_{ij} = (Y_{ij} + Y_{ij}^c)^* \frac{|\mathbf{V}_i|^2}{|T_{ij}|^2} - Y_{ij}^* \frac{\mathbf{V}_i \mathbf{V}_j^*}{T_{ij}} \quad \forall ij \in \mathcal{E} \\
 & \tilde{\mathbf{S}}_{ij} = (Y_{ij} + Y_{ji}^c)^* |\mathbf{V}_j|^2 - Y_{ij}^* \frac{\mathbf{V}_i^* \mathbf{V}_j}{T_{ij}^*} \quad \forall ij \in \mathcal{E} \\
 & |\vec{\mathbf{S}}_{ij}|, |\tilde{\mathbf{S}}_{ij}| \leq \bar{s}_{ij} \quad \forall ij \in \mathcal{E} \quad \text{Line thermal limit} \\
 & \underline{v}_i \leq |\boxed{\mathbf{V}_i}| \leq \bar{v}_i \quad \text{Voltage (complex)} \quad \forall i \in \mathcal{N} \\
 & \underline{\theta}_{ij} \leq \boxed{\theta_i} - \boxed{\theta_j} \leq \bar{\theta}_{ij} \quad \forall ij \in \mathcal{E} \\
 & \underline{p}_i^g \leq \boxed{\mathbf{p}_i^g} \leq \bar{p}_i^g \quad \text{Active (real) power generation} \quad \forall i \in \mathcal{N} \\
 & \underline{q}_i^g \leq \boxed{\mathbf{q}_i^g} \leq \bar{q}_i^g \quad \text{Reactive (imaginary) power generation} \quad \forall i \in \mathcal{N}
 \end{aligned}$$

Load (power demand)

Power flow equations

AC-OPF (Optimal Power Flow)

$$\begin{aligned} \min \quad & \sum_{i \in \mathcal{N}} c_i p_i^g \\ \text{s.t.} \quad & \mathbf{S}_i^g - \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{S}}_e - \sum_{e \in \mathcal{E}_i^-} \tilde{\mathbf{S}}_e - (Y_i^s)^* |\mathbf{V}_i|^2 = \boxed{\mathbf{S}_i^d} \quad \forall i \in \mathcal{N} \end{aligned}$$

Load (power demand)

Supply-demand balance

$$\begin{aligned} \vec{\mathbf{S}}_{ij} &= (Y_{ij} + Y_{ij}^c)^* \frac{|\mathbf{V}_i|^2}{|T_{ij}|^2} - Y_{ij}^* \frac{\mathbf{V}_i \mathbf{V}_j^*}{T_{ij}} \\ \tilde{\mathbf{S}}_{ij} &= (Y_{ij} + Y_{ij}^c)^* \frac{|\mathbf{V}_j|^2}{|T_{ij}|^2} - Y_{ij}^* \frac{\mathbf{V}_i \mathbf{V}_j^*}{T_{ij}} \end{aligned} \quad \forall ij \in \mathcal{E}$$

**Nonlinear, nonconvex, valuable
for power system operations**

Power flow equations

$$|\vec{\mathbf{S}}_{ij}|, |\tilde{\mathbf{S}}_{ij}| \leq \bar{s}_{ij} \quad \forall ij \in \mathcal{E}$$

Line thermal limit

$$\underline{v}_i \leq |\boxed{\mathbf{V}_i}| \leq \bar{v}_i \quad \text{Voltage (complex)} \quad \forall i \in \mathcal{N}$$

$$\underline{\theta}_{ij} \leq \boxed{\theta_i} - \boxed{\theta_j} \leq \bar{\theta}_{ij} \quad \forall ij \in \mathcal{E}$$

$$\underline{p}_i^g \leq \boxed{p_i^g} \leq \bar{p}_i^g \quad \text{Active (real) power generation} \quad \forall i \in \mathcal{N}$$

$$\underline{q}_i^g \leq \boxed{q_i^g} \leq \bar{q}_i^g \quad \text{Reactive (imaginary) power generation} \quad \forall i \in \mathcal{N}$$

- ▶ A feasible optimization proxy (efficiently generates solutions of optimization problems)
 - real-time operations
 - simulations
 - multi-stage optimization problems
- ▶ ...for the dual of a convex relaxation of AC-OPF
 - valid lower bound for AC-OPF
 - verify quality of past solutions, primal proxy solutions...
- ▶ ...without needing ground truth solutions as training targets (self-supervision)
 - avoids solving cost
 - results in better solutions in some past works (e.g., [1])

- ▶ "Most convex optimization problems in practice are conic-representable"
- ▶ Addresses specific structures
 - e.g., success of interior point method

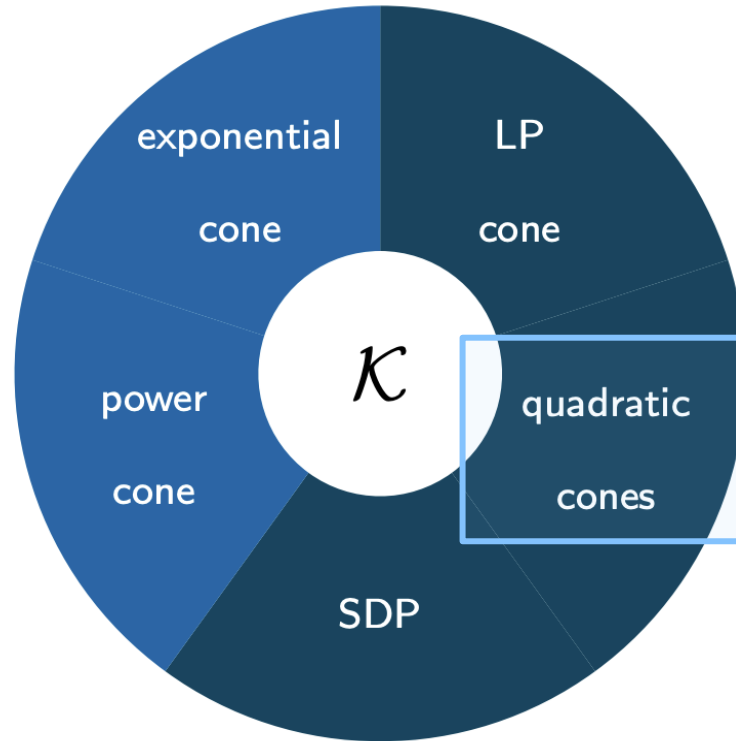


Figure 1. Classification of convex cones. [2]

Second Order Cone Relaxation

$$\begin{aligned}
 \min \quad & \sum_{i \in \mathcal{N}} c_i \mathbf{p}_i^g \\
 \text{s.t.} \quad & \mathbf{p}_i^g - \boxed{\mathbf{p}_i^d} - g_i^s \mathbf{w}_i = \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{p}}_e - \sum_{e \in \mathcal{E}_i^-} \vec{\mathbf{p}}_e \quad \forall i \in \mathcal{N} \\
 & \mathbf{q}_i^g - \boxed{\mathbf{q}_i^d} + b_i^s \mathbf{w}_i = \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{q}}_e - \sum_{e \in \mathcal{E}_i^-} \vec{\mathbf{q}}_e \quad \forall i \in \mathcal{N} \\
 & \vec{\mathbf{p}}_{ij} = \vec{\gamma}_{ij}^{p,w} \mathbf{w}_i + \vec{\gamma}_{ij}^{p,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{p,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{p}}_{ij} = \vec{\gamma}_{ij}^{p,w} \mathbf{w}_j + \vec{\gamma}_{ij}^{p,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{p,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{q}}_{ij} = \vec{\gamma}_{ij}^{q,w} \mathbf{w}_i + \vec{\gamma}_{ij}^{q,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{q,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{q}}_{ij} = \vec{\gamma}_{ij}^{q,w} \mathbf{w}_j + \vec{\gamma}_{ij}^{q,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{q,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E} \\
 & (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E} \\
 & \tan(\underline{\theta}_{ij}) \boxed{\mathbf{w}_{ij}^{\text{re}}} \leq \boxed{\mathbf{w}_{ij}^{\text{im}}} \leq \tan(\bar{\theta}_{ij}) \mathbf{w}_{ij}^{\text{re}} \quad \forall ij \in \mathcal{E} \\
 & \left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4 \quad \forall ij \in \mathcal{E} \\
 & \underline{v}_i^2 \leq \boxed{\mathbf{w}_i} \leq \bar{v}_i^2 \quad \forall i \in \mathcal{N} \\
 & \underline{\mathbf{p}}_i^g \leq \boxed{\mathbf{p}_i^g} \leq \bar{\mathbf{p}}_i^g \quad \forall i \in \mathcal{N} \\
 & \underline{\mathbf{q}}_i^g \leq \boxed{\mathbf{q}_i^g} \leq \bar{\mathbf{q}}_i^g \quad \forall i \in \mathcal{N}
 \end{aligned}$$

Supply-demand balance

Power flow equations

Line thermal limit

Change of variables

$$\begin{aligned}
 \mathbf{w}_i &= \mathbf{v}_i^2 & \forall i \in \mathcal{N} \\
 \mathbf{w}_{ij}^{\text{re}} &= \mathbf{v}_i \mathbf{v}_j \cos(\theta_i - \theta_j) & \forall ij \in \mathcal{E} \\
 \mathbf{w}_{ij}^{\text{im}} &= \mathbf{v}_i \mathbf{v}_j \sin(\theta_i - \theta_j) & \forall ij \in \mathcal{E}
 \end{aligned}$$

Second Order Cone Relaxation

$$\begin{aligned}
 \min \quad & \sum_{i \in \mathcal{N}} c_i \mathbf{p}_i^g \\
 \text{s.t.} \quad & \mathbf{p}_i^g - \boxed{\mathbf{p}_i^d} - g_i^s \mathbf{w}_i = \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{p}}_e - \sum_{e \in \mathcal{E}_i^-} \vec{\mathbf{p}}_e \quad \forall i \in \mathcal{N} \\
 & \mathbf{q}_i^g - \boxed{\mathbf{q}_i^d} + b_i^s \mathbf{w}_i = \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{q}}_e - \sum_{e \in \mathcal{E}_i^-} \vec{\mathbf{q}}_e \quad \forall i \in \mathcal{N} \\
 & \vec{\mathbf{p}}_{ij} = \vec{\gamma}_{ij}^{p,w} \mathbf{w}_i + \vec{\gamma}_{ij}^{p,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{p,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{p}}_{ij} = \vec{\gamma}_{ij}^{p,w} \mathbf{w}_j + \vec{\gamma}_{ij}^{p,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{p,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{q}}_{ij} = \vec{\gamma}_{ij}^{q,w} \mathbf{w}_i + \vec{\gamma}_{ij}^{q,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{q,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{q}}_{ij} = \vec{\gamma}_{ij}^{q,w} \mathbf{w}_j + \vec{\gamma}_{ij}^{q,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{q,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E} \\
 & (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E} \\
 & \tan(\underline{\theta}_{ij}) \boxed{\mathbf{w}_{ij}^{\text{re}}} \leq \boxed{\mathbf{w}_{ij}^{\text{im}}} \leq \tan(\bar{\theta}_{ij}) \mathbf{w}_{ij}^{\text{re}} \quad \forall ij \in \mathcal{E} \\
 & \left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4 \quad \forall ij \in \mathcal{E} \\
 & \underline{v}_i^2 \leq \boxed{\mathbf{w}_i} \leq \bar{v}_i^2 \quad \forall i \in \mathcal{N} \\
 & \underline{\mathbf{p}}_i^g \leq \boxed{\mathbf{p}_i^g} \leq \bar{\mathbf{p}}_i^g \quad \forall i \in \mathcal{N} \\
 & \underline{\mathbf{q}}_i^g \leq \boxed{\mathbf{q}_i^g} \leq \bar{\mathbf{q}}_i^g \quad \forall i \in \mathcal{N}
 \end{aligned}$$

???

[3]



Second Order Cone Formulation

$$(\bar{s}_{ij}, \vec{p}_{ij}, \vec{q}_{ij}) \in \mathcal{Q}^3$$

$$\forall ij \in \mathcal{E}$$

$$(\bar{s}_{ij}, \tilde{p}_{ij}, \tilde{q}_{ij}) \in \mathcal{Q}^3$$

$$\forall ij \in \mathcal{E}$$

$$\left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}}\right) \in \mathcal{Q}_r^4$$

$$\forall ij \in \mathcal{E}$$

- ▶ Let us define second order cone and rotated second order cone

$$\mathcal{Q}^n$$

$$\{x \in \mathbb{R}^n \mid x_1 \geq \sqrt{x_2^2 + \dots + x_n^2}\}$$

$$\mathcal{Q}_r^n$$

$$\{x \in \mathbb{R}^n \mid 2x_1x_2 \geq x_3^2 + \dots + x_n^2, x_1, x_2 \geq 0\}$$

Second Order Cone Formulation

$$\begin{array}{ll}
 |\vec{\mathbf{S}}_{ij}|, |\tilde{\mathbf{S}}_{ij}| \leq \bar{s}_{ij} & \longrightarrow \begin{array}{l} (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \\ (\bar{s}_{ij}, \tilde{\mathbf{p}}_{ij}, \tilde{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \end{array} & \begin{array}{l} \forall ij \in \mathcal{E} \\ \forall ij \in \mathcal{E} \end{array} \\
 (\mathbf{w}_{ij}^{\text{re}})^2 + (\mathbf{w}_{ij}^{\text{im}})^2 \leq \mathbf{w}_i \mathbf{w}_j & \longrightarrow \left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4 & \forall ij \in \mathcal{E}
 \end{array}$$

- Let us define second order cone and rotated second order cone

$$\begin{array}{ll}
 \mathcal{Q}^n & \{x \in \mathbb{R}^n \mid x_1 \geq \sqrt{x_2^2 + \dots + x_n^2}\} \\
 \mathcal{Q}_r^n & \{x \in \mathbb{R}^n \mid 2x_1x_2 \geq x_3^2 + \dots + x_n^2, x_1, x_2 \geq 0\}
 \end{array}$$

Second Order Cone Formulation

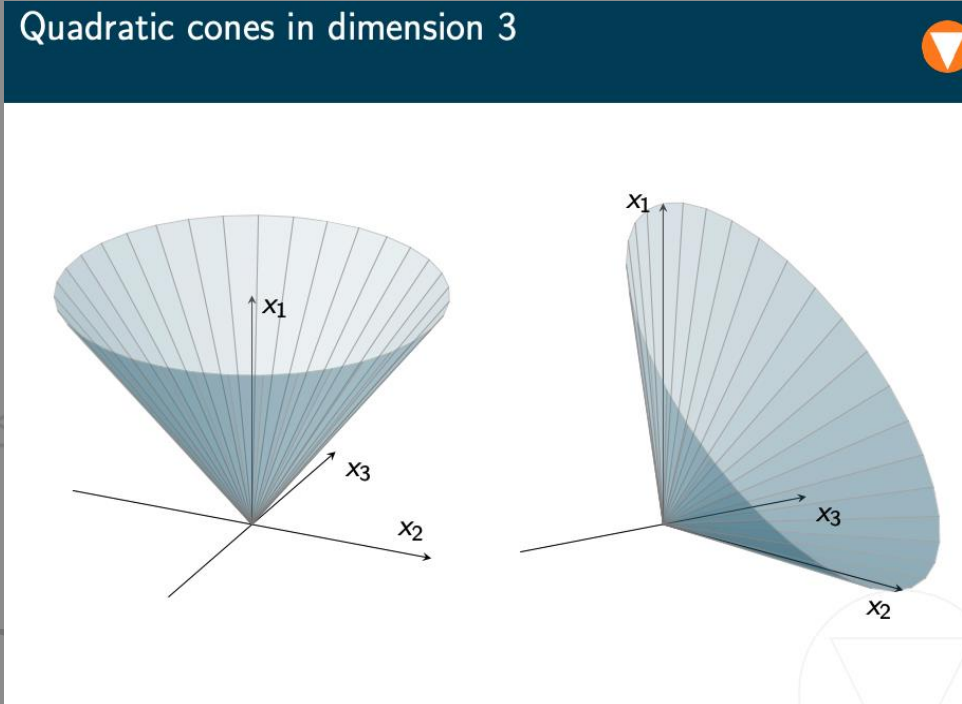


Figure 2. Second order cones in dimension 3. [2]

► Let us define second

order cone

$$Q^n = \{x \in \mathbb{R}^n \mid x_1 \geq \sqrt{x_2^2 + \dots + x_n^2}\}$$
$$Q_r^n = \{x \in \mathbb{R}^n \mid 2x_1x_2 \geq x_3^2 + \dots + x_n^2, x_1, x_2 \geq 0\}$$

► We call the relaxation SOC-OPF (“second order cone”).

Building the Dual of SOC-OPF

$$\begin{aligned}
 \min \quad & \sum_{i \in \mathcal{N}} c_i \mathbf{p}_i^g \\
 \text{s.t.} \quad & \mathbf{p}_i^g - \boxed{\mathbf{p}_i^d} - g_i^s \mathbf{w}_i = \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{p}}_e - \sum_{e \in \mathcal{E}_i^-} \vec{\mathbf{p}}_e \quad \forall i \in \mathcal{N} \\
 & \mathbf{q}_i^g - \boxed{\mathbf{q}_i^d} + b_i^s \mathbf{w}_i = \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{q}}_e - \sum_{e \in \mathcal{E}_i^-} \vec{\mathbf{q}}_e \quad \forall i \in \mathcal{N} \\
 & \vec{\mathbf{p}}_{ij} = \vec{\gamma}_{ij}^{p,w} \mathbf{w}_i + \vec{\gamma}_{ij}^{p,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{p,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{p}}_{ij} = \vec{\gamma}_{ij}^{p,w} \mathbf{w}_j + \vec{\gamma}_{ij}^{p,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{p,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{q}}_{ij} = \vec{\gamma}_{ij}^{q,w} \mathbf{w}_i + \vec{\gamma}_{ij}^{q,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{q,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & \vec{\mathbf{q}}_{ij} = \vec{\gamma}_{ij}^{q,w} \mathbf{w}_j + \vec{\gamma}_{ij}^{q,r} \mathbf{w}_{ij}^{\text{re}} + \vec{\gamma}_{ij}^{q,i} \mathbf{w}_{ij}^{\text{im}} \quad \forall ij \in \mathcal{E} \\
 & (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E} \\
 & (\bar{s}_{ij}, \vec{\mathbf{p}}_{ij}, \vec{\mathbf{q}}_{ij}) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E} \\
 & \tan(\underline{\theta}_{ij}) \mathbf{w}_{ij}^{\text{re}} \leq \boxed{\mathbf{w}_{ij}^{\text{im}}} \leq \tan(\bar{\theta}_{ij}) \mathbf{w}_{ij}^{\text{re}} \quad \forall ij \in \mathcal{E} \\
 & \left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4 \quad \forall ij \in \mathcal{E} \\
 & \underline{v}_i^2 \leq \boxed{\mathbf{w}_i} \leq \bar{v}_i^2 \quad \forall i \in \mathcal{N} \\
 & \underline{\mathbf{p}}_i^g \leq \boxed{\mathbf{p}_i^g} \leq \bar{\mathbf{p}}_i^g \quad \forall i \in \mathcal{N} \\
 & \underline{\mathbf{q}}_i^g \leq \boxed{\mathbf{q}_i^g} \leq \bar{\mathbf{q}}_i^g \quad \forall i \in \mathcal{N}
 \end{aligned}$$

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Conic Dual of SOC-OPF

$$\begin{aligned}
 \max_{\lambda, \mu, \nu} \quad & \sum_{i \in \mathcal{N}} \left(\underline{p}_i^d \lambda_i^p + \underline{q}_i^d \lambda_i^q + \underline{p}_i^g \mu_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \mu_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \mu_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s) \\
 \text{s.t.} \quad & \lambda_i^p + \mu_i^{pg} - \bar{\mu}_i^{pg} = c_i & \forall i \in \mathcal{N} \\
 & \lambda_i^q + \mu_i^{qg} - \bar{\mu}_i^{qg} = 0 & \forall i \in \mathcal{N} \\
 & -\lambda_i^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_i^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -g_i^s \lambda_i^p + b_i^s \lambda_i^q + \sum_{e \in \mathcal{E}_i^+} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^f}{\sqrt{2}} \right) + \sum_{e \in \mathcal{E}_i^-} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^t}{\sqrt{2}} \right) + \mu_i^w - \bar{\mu}_i^w = 0 & \forall i \in \mathcal{N} \\
 & \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q - \tan(\theta_{ij}) \mu_{ij}^\theta + \tan(\bar{\theta}_{ij}) \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{re}} = 0 & \forall ij \in \mathcal{E} \\
 & \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \mu_{ij}^\theta - \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{im}} = 0 & \forall ij \in \mathcal{E} \\
 & \mu_i^{pg}, \bar{\mu}_i^{pg}, \mu_i^{qg}, \bar{\mu}_i^{qg}, \mu_i^w, \bar{\mu}_i^w, \mu_i^\theta, \bar{\mu}_i^\theta \geq 0 \\
 & \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3 & \forall ij \in \mathcal{E} \\
 & \omega_{ij} = (\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{\text{re}}, \omega_{ij}^{\text{im}}) \in \mathcal{Q}_r^4, & \forall ij \in \mathcal{E}
 \end{aligned}$$

$$\begin{aligned}
 (\bar{s}_{ij}, \bar{\mathbf{p}}_{ij}, \bar{\mathbf{q}}_{ij}) &\in \mathcal{Q}^3 \\
 (\bar{s}_{ij}, \bar{\mathbf{p}}_{ij}, \bar{\mathbf{q}}_{ij}) &\in \mathcal{Q}^3
 \end{aligned}$$

$$\left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4$$

Similar to duality of linear programs



Conic Dual of SOC-OPF

$$\max_{\lambda, \mu, \nu} \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \underline{\mu}_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \underline{\mu}_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s)$$

$$s.t. \lambda_i^p + \underline{\mu}_i^{pg} - \bar{\mu}_i^{pg} = c_i \quad \forall i \in \mathcal{N}$$

$$\lambda_i^q + \underline{\mu}_i^{qg} - \bar{\mu}_i^{qg} = 0 \quad \forall i \in \mathcal{N}$$

$$-\lambda_i^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 \quad \forall ij \in \mathcal{E}$$

$$-\lambda_i^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 \quad \forall ij \in \mathcal{E}$$

$$-\lambda_j^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 \quad \forall ij \in \mathcal{E}$$

$$-\lambda_j^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 \quad \forall ij \in \mathcal{E}$$

How to learn to solve this problem?

$$-g_i^s \lambda_i^p + b_i^s \lambda_i^q + \sum_{e \in \mathcal{E}_i^+} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^f}{\sqrt{2}} \right) + \sum_{e \in \mathcal{E}_i^-} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^t}{\sqrt{2}} \right) + \underline{\mu}_i^w - \bar{\mu}_i^w = 0 \quad \forall i \in \mathcal{N}$$

$$\bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q - \tan(\theta_{ij}) \underline{\mu}_{ij}^\theta + \tan(\bar{\theta}_{ij}) \bar{\mu}_{ij}^\theta + \omega_{ij}^{re} = 0 \quad \forall ij \in \mathcal{E}$$

$$\bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \underline{\mu}_{ij}^\theta - \bar{\mu}_{ij}^\theta + \omega_{ij}^{im} = 0 \quad \forall ij \in \mathcal{E}$$

$$\underline{\mu}^{pg}, \bar{\mu}^{pg}, \underline{\mu}^{qg}, \bar{\mu}^{qg}, \underline{\mu}^w, \bar{\mu}^w, \underline{\mu}^\theta, \bar{\mu}^\theta \geq 0$$

$$\bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3$$

$$\omega_{ij} = (\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{re}, \omega_{ij}^{im}) \in \mathcal{Q}_r^4,$$

$$(\bar{s}_{ij}, \bar{\mathbf{p}}_{ij}, \bar{\mathbf{q}}_{ij}) \in \mathcal{Q}^3$$

$$(\bar{s}_{ij}, \bar{\mathbf{p}}_{ij}, \bar{\mathbf{q}}_{ij}) \in \mathcal{Q}^3$$

$$(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{re}, \mathbf{w}_{ij}^{im}) \in \mathcal{Q}_r^4$$

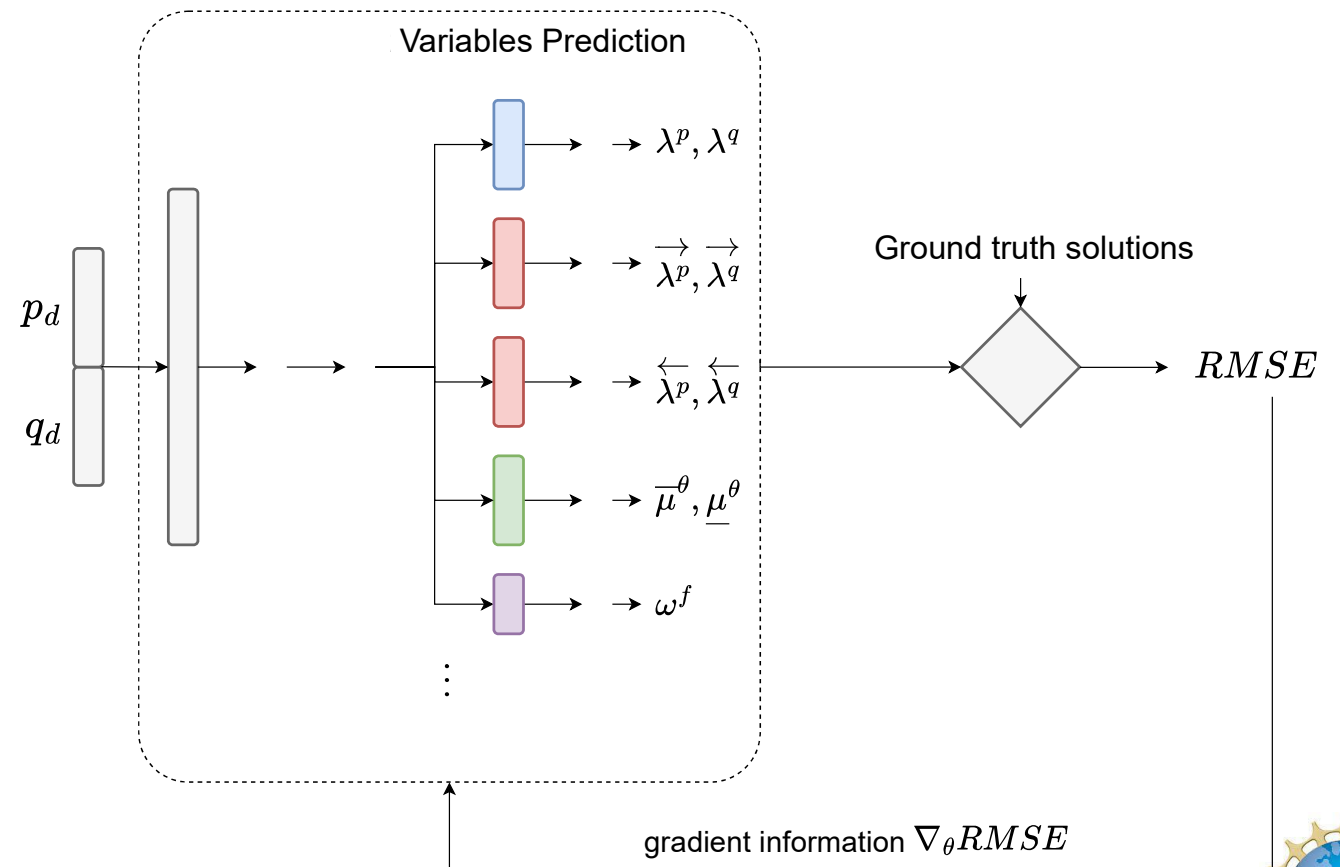
Similar to duality of linear programs



Neural Network Proxy

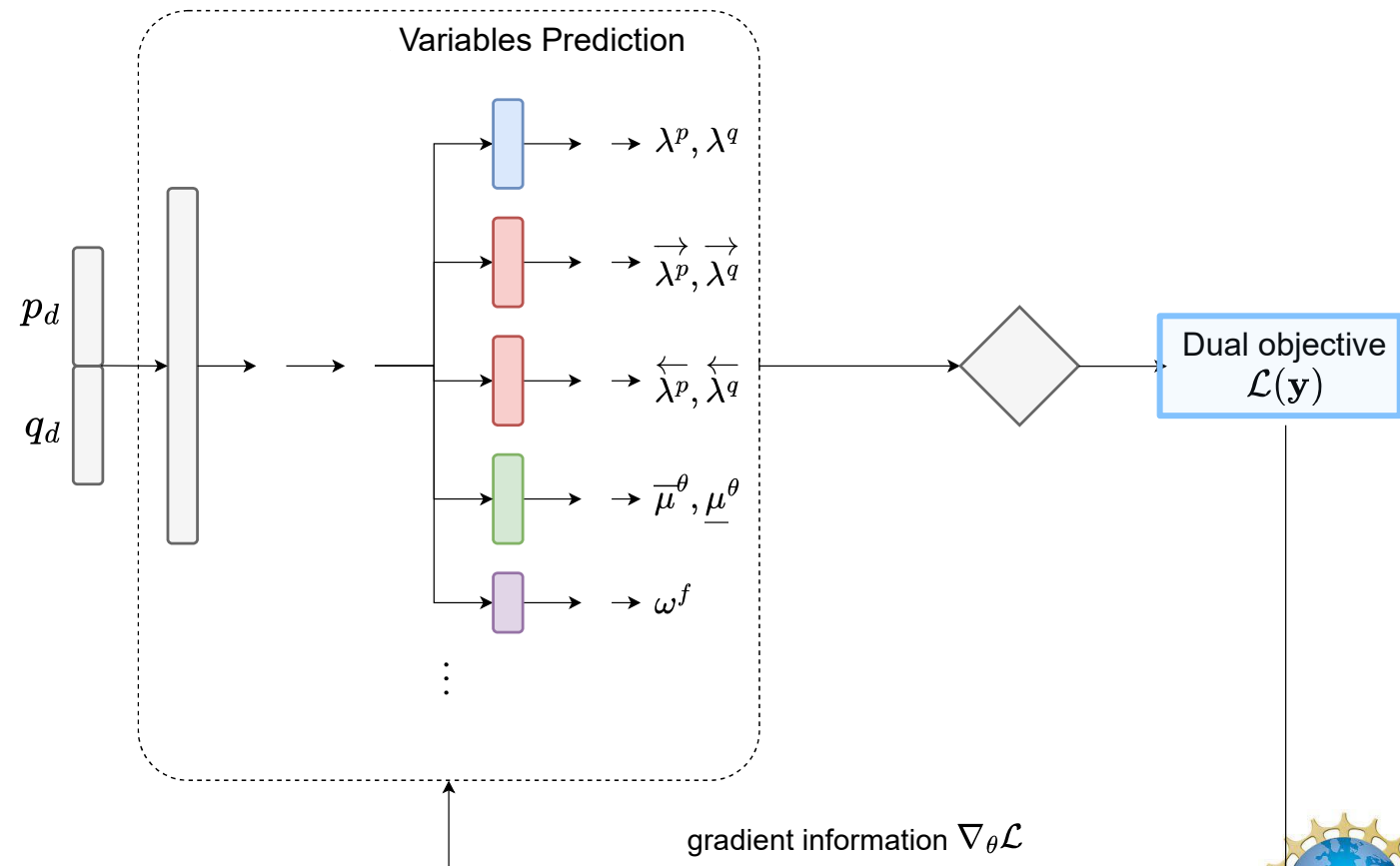
- A **supervised** proxy that learns to imitate solver output given problem instance parameters.

– Feasibility?



Neural Network Proxy

- A **self-supervised** proxy that learns to produce good solutions without ground truth solutions.
 - How to enforce the constraints?
 - Use properties of the dual!



Utilizing Properties of the Dual

$$\begin{aligned}
 \max_{\lambda, \mu, \nu} \quad & \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \underline{\mu}_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \underline{\mu}_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s) \\
 \text{s.t.} \quad & \lambda_i^p + \underline{\mu}_i^{pg} - \bar{\mu}_i^{pg} = c_i & \forall i \in \mathcal{N} \\
 & \lambda_i^q + \underline{\mu}_i^{qg} - \bar{\mu}_i^{qg} = 0 & \forall i \in \mathcal{N} \\
 & -\lambda_i^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_i^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -g_i^s \lambda_i^p + b_i^s \lambda_i^q + \sum_{e \in \mathcal{E}_i^+} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^f}{\sqrt{2}} \right) + \sum_{e \in \mathcal{E}_i^-} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^t}{\sqrt{2}} \right) + \underline{\mu}_i^w - \bar{\mu}_i^w = 0 & \forall i \in \mathcal{N} \\
 & \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q - \tan(\theta_{ij}) \underline{\mu}_{ij}^\theta + \tan(\bar{\theta}_{ij}) \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{re}} = 0 & \forall ij \in \mathcal{E} \\
 & \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \underline{\mu}_{ij}^\theta - \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{im}} = 0 & \forall ij \in \mathcal{E} \\
 & \underline{\mu}^{pg}, \bar{\mu}^{pg}, \underline{\mu}^{qg}, \bar{\mu}^{qg}, \underline{\mu}^w, \bar{\mu}^w, \underline{\mu}^\theta, \bar{\mu}^\theta \geq 0 \\
 & \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3 & \forall ij \in \mathcal{E} \\
 & \omega_{ij} = (\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{\text{re}}, \omega_{ij}^{\text{im}}) \in \mathcal{Q}_r^4, & \forall ij \in \mathcal{E}
 \end{aligned}$$



Utilizing Properties of the Dual

$$\begin{aligned} \max_{\lambda, \mu, \nu} \quad & \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} \right) \\ \text{s.t.} \quad & \lambda_i^p + \underline{\mu}_i^{pg} - \bar{\mu}_i^{pg} = c_i \end{aligned}$$

At most one should be nonzero
Otherwise, suboptimal!

*Alternatively, due to
complementary slackness...

Primal constraint

$$\underline{p}_i^g \leq p_i^g \leq \bar{p}_i^g$$

- Feasibility of equality constraints
 - No loss of optimality

Utilizing Properties of the Dual

$$\begin{aligned}
 \max_{\lambda, \mu, \nu} \quad & \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \underline{\mu}_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \underline{\mu}_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s) \\
 \text{s.t.} \quad & \lambda_i^p + \underline{\mu}_i^{pg} - \bar{\mu}_i^{pg} = c_i & \forall i \in \mathcal{N} \\
 & \lambda_i^q + \underline{\mu}_i^{qg} - \bar{\mu}_i^{qg} = 0 & \forall i \in \mathcal{N} \\
 & -\lambda_i^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_i^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -g_i^s \lambda_i^p + b_i^s \lambda_i^q + \sum_{e \in \mathcal{E}_i^+} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^f}{\sqrt{2}} \right) + \sum_{e \in \mathcal{E}_i^-} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^t}{\sqrt{2}} \right) + \underline{\mu}_i^w - \bar{\mu}_i^w = 0 & \forall i \in \mathcal{N} \\
 & \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q - \tan(\theta_{ij}) \underline{\mu}_{ij}^\theta + \tan(\bar{\theta}_{ij}) \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{re}} = 0 & \forall ij \in \mathcal{E} \\
 & \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \underline{\mu}_{ij}^\theta - \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{im}} = 0 & \forall ij \in \mathcal{E} \\
 & \underline{\mu}^{pg}, \bar{\mu}^{pg}, \underline{\mu}^{qg}, \bar{\mu}^{qg}, \underline{\mu}^w, \bar{\mu}^w, \underline{\mu}^\theta, \bar{\mu}^\theta \geq 0 \\
 & \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3 & \forall ij \in \mathcal{E} \\
 & \omega_{ij} = (\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{\text{re}}, \omega_{ij}^{\text{im}}) \in \mathcal{Q}_r^4, & \forall ij \in \mathcal{E}
 \end{aligned}$$



Utilizing Properties of the Dual

$$\max_{\lambda, \mu, \nu} \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \mu_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \mu_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \mu_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s)$$

$$\vec{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3$$

$$\mathcal{Q}^n \quad \{x \in \mathbb{R}^n \mid x_1 \geq \sqrt{x_2^2 + \dots + x_n^2}\}$$

Must be =
Otherwise, suboptimal!

- Feasibility of the conic constraints
 - One fewer variable to predict
 - No loss of optimality

Utilizing Properties of the Dual

$$\begin{aligned}
 \max_{\lambda, \mu, \nu} \quad & \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \underline{\mu}_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \underline{\mu}_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s) \\
 \text{s.t.} \quad & \lambda_i^p + \underline{\mu}_i^{pg} - \bar{\mu}_i^{pg} = c_i & \forall i \in \mathcal{N} \\
 & \lambda_i^q + \underline{\mu}_i^{qg} - \bar{\mu}_i^{qg} = 0 & \forall i \in \mathcal{N} \\
 & -\lambda_i^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_i^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 & \forall ij \in \mathcal{E} \\
 & -\lambda_j^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 & \forall ij \in \mathcal{E} \\
 & -g_i^s \lambda_i^p + b_i^s \lambda_i^q + \sum_{e \in \mathcal{E}_i^+} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^f}{\sqrt{2}} \right) + \sum_{e \in \mathcal{E}_i^-} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^t}{\sqrt{2}} \right) + \underline{\mu}_i^w - \bar{\mu}_i^w = 0 & \forall i \in \mathcal{N} \\
 & \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,r} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,r} \bar{\lambda}_{ij}^q - \tan(\theta_{ij}) \underline{\mu}_{ij}^\theta + \tan(\bar{\theta}_{ij}) \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{re}} = 0 & \forall ij \in \mathcal{E} \\
 & \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \underline{\mu}_{ij}^\theta - \bar{\mu}_{ij}^\theta + \omega_{ij}^{\text{im}} = 0 & \forall ij \in \mathcal{E} \\
 & \underline{\mu}^{pg}, \bar{\mu}^{pg}, \underline{\mu}^{qg}, \bar{\mu}^{qg}, \underline{\mu}^w, \bar{\mu}^w, \underline{\mu}^\theta, \bar{\mu}^\theta \geq 0 \\
 & \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3 & \forall ij \in \mathcal{E} \\
 & \omega_{ij} = (\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{\text{re}}, \omega_{ij}^{\text{im}}) \in \mathcal{Q}_r^4, & \forall ij \in \mathcal{E}
 \end{aligned}$$



Utilizing Properties of the Dual

$$\omega_{ij} = \left(\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{\text{re}}, \omega_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4$$

$$\mathcal{Q}_r^n \quad \{x \in \mathbb{R}^n \mid 2x_1x_2 \geq x_3^2 + \dots + x_n^2, x_1, x_2 \geq 0\}$$

Primal constraint

$$\left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right) \in \mathcal{Q}_r^4$$

$$\underline{v}_i^2 \leq \mathbf{w}_i \leq \bar{v}_i^2$$

Close to 1 in practice

Must be =
Otherwise, suboptimal!

Complementary slackness*

$$\left(\frac{\mathbf{w}_i}{\sqrt{2}}, \frac{\mathbf{w}_j}{\sqrt{2}}, \mathbf{w}_{ij}^{\text{re}}, \mathbf{w}_{ij}^{\text{im}} \right)^\top \left(\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{\text{re}}, \omega_{ij}^{\text{im}} \right) = 0$$

(*strong duality can be shown)



Utilizing Properties of the Dual

$$\max_{\lambda, \mu, \nu} \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \underline{\mu}_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} + \underline{v}_i^2 \underline{\mu}_i^w - \bar{v}_i^2 \bar{\mu}_i^w \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\nu}_e^s + \bar{\nu}_e^s)$$

$$s.t. \quad \lambda_i^p + \underline{\mu}_i^{pg} - \bar{\mu}_i^{pg} = c_i \quad \forall i \in \mathcal{N}$$

$$\lambda_i^q + \underline{\mu}_i^{qg} - \bar{\mu}_i^{qg} = 0 \quad \forall i \in \mathcal{N}$$

$$-\lambda_i^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 \quad \forall ij \in \mathcal{E}$$

$$-\lambda_i^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 \quad \forall ij \in \mathcal{E}$$

$$-\lambda_j^p - \bar{\lambda}_{ij}^p + \bar{\nu}_{ij}^p = 0 \quad \forall ij \in \mathcal{E}$$

$$-\lambda_j^q - \bar{\lambda}_{ij}^q + \bar{\nu}_{ij}^q = 0 \quad \forall ij \in \mathcal{E}$$

$$-g_i^s \lambda_i^p + b_i^s \lambda_i^q + \sum_{e \in \mathcal{E}_i^+} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^f}{\sqrt{2}} \right) + \sum_{e \in \mathcal{E}_i^-} \left(\bar{\gamma}_e^{p,w} \bar{\lambda}_e^p + \bar{\gamma}_e^{q,w} \bar{\lambda}_e^q + \frac{\omega_e^t}{\sqrt{2}} \right) + \underline{\mu}_i^w - \bar{\mu}_i^w = 0 \quad \forall i \in \mathcal{N}$$

$$\bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q - \tan(\theta_{ij}) \underline{\mu}_{ij}^\theta + \tan(\bar{\theta}_{ij}) \bar{\mu}_{ij}^\theta + \omega_{ij}^{re} = 0 \quad \forall ij \in \mathcal{E}$$

$$\bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{p,i} \bar{\lambda}_{ij}^p + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \bar{\gamma}_{ij}^{q,i} \bar{\lambda}_{ij}^q + \underline{\mu}_{ij}^\theta - \bar{\mu}_{ij}^\theta + \omega_{ij}^{im} = 0 \quad \forall ij \in \mathcal{E}$$

$$\underline{\mu}^{pg}, \bar{\mu}^{pg}, \underline{\mu}^{qg}, \bar{\mu}^{qg}, \underline{\mu}^w, \bar{\mu}^w, \underline{\mu}^\theta, \bar{\mu}^\theta \geq 0$$

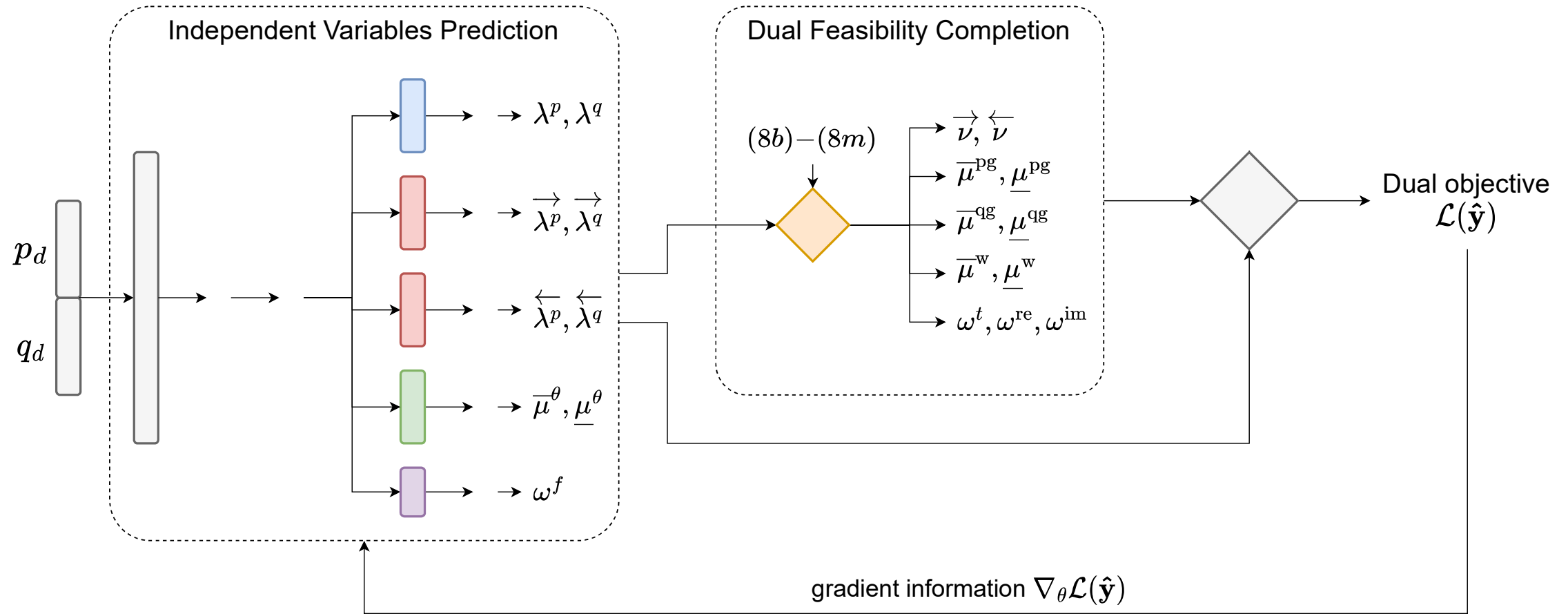
$$\bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3, \bar{\nu}_{ij} = (\bar{\nu}_{ij}^s, \bar{\nu}_{ij}^p, \bar{\nu}_{ij}^q) \in \mathcal{Q}^3 \quad \forall ij \in \mathcal{E}$$

$$\omega_{ij} = (\omega_{ij}^f, \omega_{ij}^t, \omega_{ij}^{re}, \omega_{ij}^{im}) \in \mathcal{Q}_r^4, \quad \forall ij \in \mathcal{E}$$

- All completion steps are differentiable
- The choice of variables is not unique!



The Dual Conic Proxy



Results

- Gaps of objective values compared to those obtained by an interior point solver on SOC-OPF

System	indep.	(ω^f, ω^t)	%gap – DCP		
			mean	std	max
ieee14	$\vec{\lambda}, \vec{\lambda}$	Rect.	1.17	5.50	25.20
	$\vec{\lambda}, \vec{\lambda}$	Polar	0.32	5.53	24.79
	$\vec{\nu}, \vec{\nu}$	Rect.	0.54	5.48	24.68
	$\vec{\nu}, \vec{\nu}$	Polar	0.05	5.51	24.52
ieee118	$\vec{\lambda}, \vec{\lambda}$	Rect.	0.76	0.49	2.51
	$\vec{\lambda}, \vec{\lambda}$	Polar	0.40	0.45	2.07
	$\vec{\nu}, \vec{\nu}$	Rect.	0.36	0.09	0.98
	$\vec{\nu}, \vec{\nu}$	Polar	0.08	0.15	0.57
ieee300	$\vec{\lambda}, \vec{\lambda}$	Rect.	1.47	1.46	14.86
	$\vec{\lambda}, \vec{\lambda}$	Polar	0.94	1.46	14.45
	$\vec{\nu}, \vec{\nu}$	Rect.	0.86	1.37	14.11
	$\vec{\nu}, \vec{\nu}$	Polar	0.58	1.27	13.56
pegase1354	$\vec{\lambda}, \vec{\lambda}$	Rect.	54.86	5.01	63.28
	$\vec{\lambda}, \vec{\lambda}$	Polar	54.77	5.02	63.20
	$\vec{\nu}, \vec{\nu}$	Rect.	2.14	0.23	2.66
	$\vec{\nu}, \vec{\nu}$	Polar	1.62	0.18	2.08
pegase2869	$\vec{\lambda}, \vec{\lambda}$	Rect.	67.82	4.86	75.28
	$\vec{\lambda}, \vec{\lambda}$	Polar	67.53	4.90	75.06
	$\vec{\nu}, \vec{\nu}$	Rect.	2.09	0.17	2.62
	$\vec{\nu}, \vec{\nu}$	Polar	1.57	0.18	2.11

Conclusion

- ▶ An optimization proxy that generates high-quality valid lower bounds for AC-OPF.
- ▶ Ongoing:
 - Applying the methodology to DC-OPF, risk assessment...
 - Applying the methodology to SDP relaxation of AC-OPF
 - Improve worst-case performance



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Dual Conic Proxies for AC-Optimal Power Flow



Open Source OPF Learning Tools

- ▶ AI4OPT is announcing two open-source repositories for OPF Learning
 - <https://github.com/AI4OPT>
 - **OPFGenerator**: Instance generator for reproducible and representative AC/DC/SOC-OPF datasets in Julia
 - Planned public release of pre-generated datasets
 - **ML4OPF**: Use OPFGenerator datasets for deep learning with feasibility repair layers, loss functions in PyTorch
 - Planned public release of pre-trained model checkpoints



- ▶ [1] W. Chen, Mathieu Tanneau, and Pascal Van Hentenryck, “End-to-End Feasible Optimization Proxies for Large-Scale Economic Dispatch,” *IEEE transactions on power systems*, vol. 39, no. 2, pp. 4723–4734, Mar. 2024, doi: <https://doi.org/10.1109/tpwrs.2023.3317352>.
- ▶ [2] U. Di Bologna and S. Wiese, “A gentle introduction to Continuous and Mixed-Integer Conic Programming,” 2020. Accessed: Jun. 03, 2024. [Online]. Available: <https://docs.mosek.com/slides/2020/bologna/talk.pdf>
- ▶ [3] R. A. Jabr, “Radial Distribution Load Flow Using Conic Programming,” *IEEE Transactions on Power Systems*, vol. 21, no. 3, pp. 1458–1459, Aug. 2006, doi: <https://doi.org/10.1109/tpwrs.2006.879234>.

Thank you

